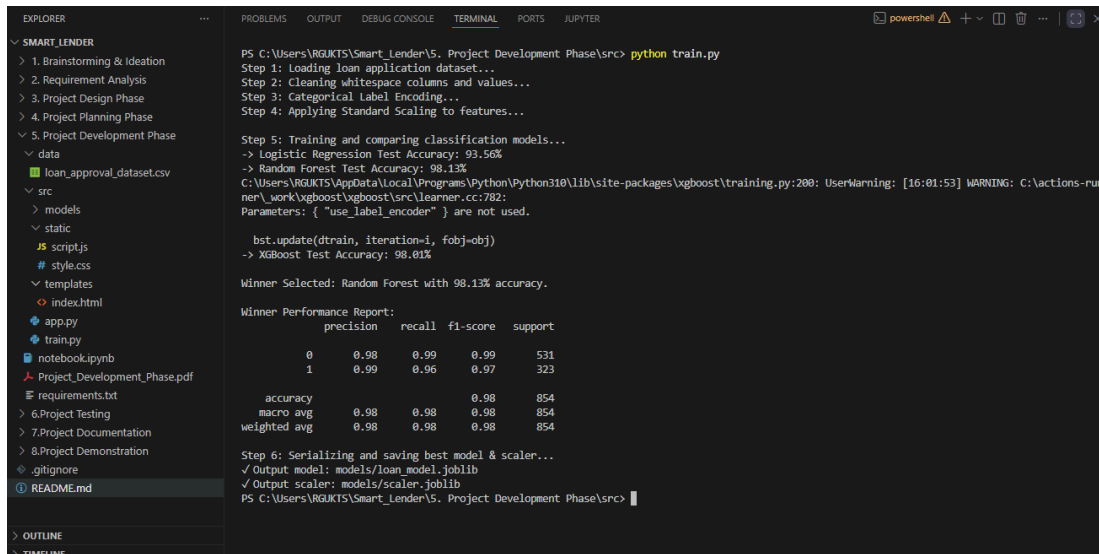


Phase 8: Project Demonstration

This section demonstrates the Smart Lender application running end-to-end. Note: this page is a ready-to-fill template — please insert your own screenshots (taken after running the app on your machine) in place of each placeholder box below before adding this file to the repository.

Terminal Execution

Insert a screenshot of the terminal showing train.py running successfully (model comparison + accuracy output) and app.py starting the Flask server (e.g., "Running on <http://127.0.0.1:5000>")



```
PS C:\Users\RGUKTS\Smart_Lender\S. Project Development Phase\src> python train.py
Step 1: Loading loan application dataset...
Step 2: Cleaning whitespace columns and values...
Step 3: Categorical Label Encoding...
Step 4: Applying Standard Scaling to features...

Step 5: Training and comparing classification models...
-> Logistic Regression Test Accuracy: 93.56%
-> Random Forest Test Accuracy: 98.13%
C:\Users\RGUKTS\AppData\Local\Programs\Python\Python310\lib\site-packages\xgboost\training.py:208: UserWarning: [16:01:53] WARNING: C:\actions-run
ner\work\xgboost\xgboost\src\learner.cc:782:
Parameters: { "use_label_encoder" } are not used.

  bst.update(dtrain, iteration=i, fobj=obj)
-> XGBoost Test Accuracy: 98.01%

Winner Selected: Random Forest with 98.13% accuracy.

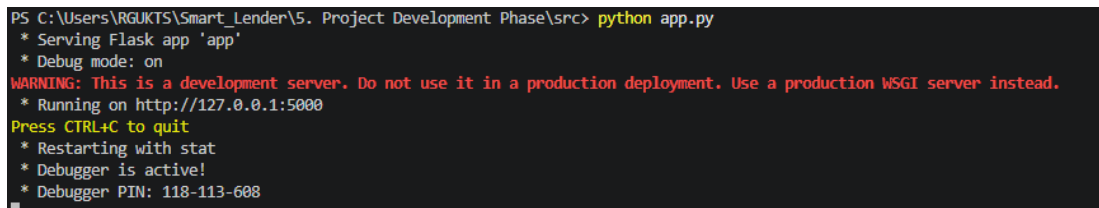
Winner Performance Report:
      precision    recall  f1-score   support

         0       0.98       0.99       0.99         531
         1       0.99       0.96       0.97         323

 accuracy
macro avg       0.98       0.98       0.98         854
weighted avg       0.98       0.98       0.98         854

Step 6: Serializing and saving best model & scaler...
✓ Output model: models/loan_model.joblib
✓ Output scaler: models/scaler.joblib
PS C:\Users\RGUKTS\Smart_Lender\S. Project Development Phase\src>
```

Python train.py output



```
PS C:\Users\RGUKTS\Smart_Lender\S. Project Development Phase\src> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 118-113-608
```

Python app.py output

Web Interface

Insert screenshots of the running web application, showing the applicant details form and the prediction result screen for both an Approved and a Rejected case.

SmartLender

Applicant Details

Education

Graduate

Self Employed

No

No. of Dependents

eg : 2

Loan Term (Months)

eg : 12

Annual Income (₹)

eg : 50,000

Loan Amount (₹)

eg : 1,00,000

CIBIL Score: 300

Don't know your CIBIL score? Check it on your bank app or [CIBIL website](#).

Asset Values (₹)

Residential Assets

eg : 20,000

Commercial Assets

eg : 1,00,000

Luxury Assets

eg : 50,000

Liquid Bank Assets

eg : 1,00,000

Predict Approval



Fill the form and click **Predict Approval** to see your loan result here.

Loan application form

SmartLender

Applicant Details

Education

Graduate

Self Employed

Yes

No. of Dependents

4

Loan Term (Months)

50

Annual Income (₹)

30000

Loan Amount (₹)

500000

CIBIL Score: 553

Don't know your CIBIL score? Check it on your bank app or [CIBIL website](#).

Asset Values (₹)

Residential Assets

40000

Commercial Assets

100000

Luxury Assets

200000

Liquid Bank Assets

50000

Predict Approval



Loan Approved!

78.0% Approval Probability

Check Another Application

Prediction result (Approved case)

SmartLender

Applicant Details

Education

Graduate

Self Employed

No

No. of Dependents

1

Loan Term (Months)

24

Annual Income (₹)

10000

Loan Amount (₹)

100000

CIBIL Score: 300

Don't know your CIBIL score? Check it on your bank app or [CIBIL website](#).

Asset Values (₹)

Residential Assets

10000

Commercial Assets

20000

Luxury Assets

0

Liquid Bank Assets

0

Predict Approval



Loan Rejected

93.0% Approval Probability

Check Another Application

Prediction result (Rejected case)

Demo Video Link

Record a short (2–3 minute) screen walkthrough showing the application being run from the terminal and used in the browser, upload it (e.g., to Google Drive or YouTube as Unlisted), and paste the link below.

[Smart_lender.mp4](#)